Phase 21: End-to-End Business Analyst Case Study

Case Study Title: Scaling Storage Governance & Eliminating Redundant Cloud Costs via Hybrid Deduplication

Project: Duplicate Bin Ecosystem

Candidate Persona: Senior Technical Business Analyst / Product Owner

Target Audience: Hiring Managers, Principal BAs, Interview Panels

1. Business Context & Strategic Drivers

In digital-first organizations and creative media enterprises, unstructured file data expands at an estimated 30% to 50% compound annual growth rate. Creative media agencies, architectural studios, and software engineering teams generate massive repositories of digital assets—RAW photos, 4K video rushes, design exports, and project staging builds.

Because traditional operating systems lack automated deduplication intelligence, redundant files rapidly consume between **15% and 35% of total storage capacity**. When local NVMe solid-state drives (SSDs) and corporate cloud storage quotas (Google Workspace, Microsoft OneDrive, Dropbox) breach 90% utilization, companies are forced into expensive, recurring cloud subscription tier jumps.

Duplicate Bin was initiated to solve this universal challenge by delivering an enterprise-grade, hybrid deduplication ecosystem that couples a zero-trust, privacy-first local scanning utility with a centralized SaaS management and telemetry portal.

2. The Business Problem & Pain Points

Prior to Duplicate Bin, storage cleanup across distributed workstations was characterized by severe operational bottlenecks:

1. **Human Decision Fatigue & Time Waste:** Knowledge workers spent 3 to 6 hours per month manually comparing files. Manual inspection frequently fails on identical files with differing filenames (`IMG_001.jpg` vs `Hero_banner.jpg`) or different files sharing the same name.
2. **Accidental Deletion Anxiety:** The primary psychological barrier preventing users from cleaning storage

2. **Accidental Deletion Anxiety:** The primary psychological barrier preventing users from cleaning storage

was the fear of permanently destroying unique original assets or breaking active project dependencies.

3. **Escalating Cloud Tier Costs:** Agencies paid hundreds of dollars in avoidable monthly cloud storage overage fees simply to store redundant copies of identical video exports.

4. **Lack of Central IT Governance:** IT administrators had zero fleet-wide visibility into which workstations or shared network folders were responsible for storage bloat.

3. Stakeholder Landscape

The project spanned cross-functional business, operational, and technical stakeholders:

- **Creative Prosumers & Content Creators:** Needed fast, visual side-by-side comparison, drag-and-drop ease of use, and guaranteed zero data loss.

- **Corporate IT Administrators:** Demanded headless CLI scripting (`run.py scan`), low memory footprints, and background automation without interrupting active employee work.

- **Business Owners & Agency Directors:** Focused on clear ROI, tracking aggregate gigabytes saved, and transparent software monetization (Pay-As-You-Go credits vs. team subscriptions).

- **Security & Compliance Officers:** Required strict local-only file processing guarantees ensuring zero corporate files were uploaded to external cloud environments.

4. Current State (As-Is) vs. Future State (To-Be)

AS-IS PROCESS (Manual, Siloed, High-Risk)	TO-BE PROCESS (Automated Hybrid SaaS)
• Manual inspection via OS File Explorer.	• Two-pass O(n) pre-filtering & SHA-256.
• ~100 files compared per hour.	• 10,000 files scanned in 18.7 seconds.
• Permanent deletion or messy Recycle Bin.	• Safe quarantine archive with Ctrl+Z undo
• Manual checkbox clicking per file.	• 8 automated smart selection strategies.
• Zero fleet visibility for IT teams.	• Central SaaS web portal & live telemetry
• Uncontrolled cloud tier upgrade bills.	• 15%-30% storage savings; *credit billing.*

5. My Business Analyst Contribution

Note for Interview Alignment: The implementation details below reflect the actual software system delivered

Note for Interview Alignment: The implementation details below reflect the actual software system delivered in the repository. Personal contribution to confirm based on candidate's actual involvement during interview presentation.

As the Senior Technical Business Analyst leading this initiative, the core responsibilities encompassed:

1. **Requirements Elicitation & Reverse-Engineering:** Discovered and structured 30 advanced system capabilities across core detection, performance caching, safe archiving, and cloud telemetry.
2. **Business & Functional Requirements Engineering:** Authored the formal Business Requirements Document (BRD), Functional Requirements Document (FRD), and Product Requirements Document (PRD), establishing strict Gherkin acceptance criteria.
3. **Algorithmic Business Logic Definition:** Collaborated with the Lead Architect to define the business rules for two-pass size pre-filtering and Union-Find transitive duplicate grouping.
4. **Safety Architecture Governance:** Mandated the safe archive quarantine model and instant rollback workflow (`Ctrl+Z`), eradicating the psychological barrier of data loss.
5. **API Contract & Monetization Modeling:** Defined the REST and WebSocket contracts for the FastAPI backend, specifying the 6-digit device pairing protocol and the atomic credit deduction ledger (10 GB free allowance with Pay-As-You-Go credit packs).
6. **Traceability & UAT Leadership:** Built an end-to-end Requirements Traceability Matrix (RTM) linking 24 requirements to test cases, and facilitated User Acceptance Testing with cross-functional stakeholders.

6. Core Deliverables & Artifact Library Produced

- **Business Requirements Document (BRD):** Strategic problem statement, business goals, scope boundaries, and executive sign-off matrix.
- **Functional Requirements Document (FRD):** Exhaustive specifications for all 9 core feature modules with main, alternative, and exception flows.
- **Product Requirements Document (PRD):** Product vision, target personas, MoSCoW prioritization, and go-to-market release strategy.
- **Agile User Stories & Gherkin Catalog:** 11 detailed user stories categorized under 5 epics with acceptance tests.
- **Process Flow Models:** Mermaid sequence and flowchart diagrams detailing multi-pass scanning, safe archiving, and WebSocket relay orchestration.
- **API Business Specification:** 11 REST endpoints and 2 WebSocket relays mapped to business value

- **API Business Specification:** 11 REST endpoints and 2 WebSocket relays mapped to business value and error conditions.
- **Data Dictionary & ERD:** Full cloud and desktop database schema definitions, compound cache keys, and quarantine manifest formats.
- **UAT Test Plan & Execution Log:** 10 comprehensive business scenarios executed and signed off with zero Sev-1 defects.

7. Major Project Challenges & How We Overcame Them

Challenge 1: Transitive Duplicate Group Fragmentation

- **The Issue:** Early prototype algorithms used a simple hash map key approach. When files with identical content existed across multiple directory branches with complex naming, transitive groupings ($A=B$ and $B=C$) occasionally fractured into duplicate groups or triggered key collisions.
- **BA Intervention:** Facilitated a technical spike with engineering. Specified the adoption of a **Union-Find disjoint-set data structure with path compression**, achieving mathematically guaranteed transitive clustering in $O(n \alpha(n))$ near-linear time (Architecture Decision Record DEC-001).

Challenge 2: Client Memory Spikes on Large Media Folders

- **The Issue:** Attempting to hash 4K video files by loading them directly into memory caused Out-Of-Memory (OOM) crashes on 8 GB RAM laptops.
- **BA Intervention:** Formulated Non-Functional Requirement NFR-003 mandating an **adaptive streaming chunk buffer** (4KB for small files, up to 1MB for large files). This capped client memory usage strictly below 180 MB for 10,000 files and 850 MB for 100,000 files.

Challenge 3: Balancing Data Privacy with Centralized SaaS Governance

- **The Issue:** Enterprise security officers refused to adopt any cloud solution that transmitted company file contents across the public internet.
- **BA Intervention:** Architected a **Zero-Trust Hybrid Relay model**: all file scanning, hashing, and archiving execute strictly on the local machine. Only high-level numeric telemetry (`bytes_recoverable`, `duplicates_count`, `duration_seconds`) and remote commands flow to the cloud backend.

8. Business Outcomes & Measurable Impact

Dimension	Baseline State	Measured Outcome with Duplicate Bin	Validation Source
Scan Velocity	~600 seconds (Manual)	**18.7 seconds** (10,000 files cold scan)	Verified Benchmark Table
Rescan Throughput	Full re-audit required	**4.2 seconds** (77.5% time reduction)	SQLite WAL Persistent Cache
High-Scale Performance	Crash / Unresponsive	**28.4 seconds** on 100,000 files (87.2% reduction)	Verified Benchmark Table
Data Loss Incidents	1-2% accidental deletion	**0.0% data loss** (100% quarantine recovery)	UAT Audit Manifests
Storage Reclaimed	0 GB (Cleanup deferred)	**15% to 35% typical disk recovery**	Telemetry `bytes_recoverable`
Monetization Engine	Zero revenue	Functional Pay-As-You-Go credit billing ($4.99/100GB)	FastAPI Billing Endpoints

9. Key Lessons Learned

1. **Safety Overcomes Apathy:** The breakthrough feature that drove user conversion was not raw scanning speed, but the psychological reassurance of the **Safe Archive Quarantine with instant `Ctrl+Z` undo**.
2. **Deterministic Rules Eradicate Decision Fatigue:** Users do not want to review thousands of duplicate checkmarks. Implementing **8 deterministic selection strategies** (e.g., "Keep Best Quality") transformed the user experience.
3. **Empirical Benchmarks Beat Marketing Hype:** Reconciling marketing claims ("20x faster") with empirical reality ("87% time reduction / 7.82x speedup") established credibility and prevented

3.

empirical reality ("87% time reduction / 7.82x speedup") established credibility and prevented misaligned stakeholder expectations.